

Q1. Harshad number

ANSWER

```

n=int(input("Enter a number:"))
temp=n
sum=0
while temp>0:
    digit=temp%10
    sum=sum+digit
    temp=temp//10
if n%sum==0:
    print("Harshad Number")
else:
    print("Not Harshad Number")
print("Harshad Number from 1 to 500:")
for i in range(1,501):
    temp=i
    sum=0
    while temp>0:
        digit=temp%10
        sum=sum+digit
        temp=temp//10
    if i%sum==0:
        print(i,end=" ")

```

Q2. GCD

CODE

```

a=int(input("Enter first: "))
b=int(input("Enter second: "))
while b!=0:
    temp=b
    b=a%b
    a=temp
print("GCD is",a)

```

INPUT

6
3

OUTPUT

Enter first: Enter second: GCD is 3

Q3. sum to a target value

ANSWER

```

def find_pairs(lst, target):
    seen=set()
    pairs=[]
    for num in lst:
        complement=target-num
        if complement in seen:
            pairs.append((complement, num))
            seen.add(num)
    return pairs
data = [int(x) for x in input("Enter numbers:").split()]
t= int(input("Target: "))
result= find_pairs(data, t)
print("Pairs:",result)

```

Q4. Bitwise Operators

ANSWER

```

a=0b1101
b=0o25
c=0x1A
result = (a&b)|c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)

Bitwise AND:
13= 01101
21= 10101
-----
00101 = 5

Bitwise AND
5= 00101
26= 11010
-----
11111 =31

OUTPUT:
a= 13, b= 21, c=26, result= 31
0b11111 0o37 0x1f
7 62 -32

```

Q5. binary search

CODE

```

def rotated_binary_search(arr, target):
    low= 0
    high= len(arr)-1
    while low <= high:
        mid = (low+high)//2
        if arr[mid]== target:
            return mid
        if arr[low] <= arr[mid]:
            if target >= arr[low] and target < arr[mid]:
                high = mid-1
            else:
                low =mid+1
        else:
            if target > arr[mid] and target <= arr[high]:
                low = mid+1
            else:
                high= mid-1
    return -1
data = [4,5,6,7,0,1,2]
t=int(input("Target: \n"))
result= rotated_binary_search(data, t)

```

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Bitwise AND:

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Bitwise AND

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OUTPUT:

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Q5. binary search

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                low =mid+1
        else:
            if target > arr[mid] and target ≤ arr[high]:
                low = mid+1
            else:
                high= mid-1
    return -1
data = [4,5,6,7,0,1,2]
t=int(input("Target: \n"))
result= rotated_binary_search(data, t)
print("Found at:",result)

```

OUTPUT

Target:
Found at: 4
